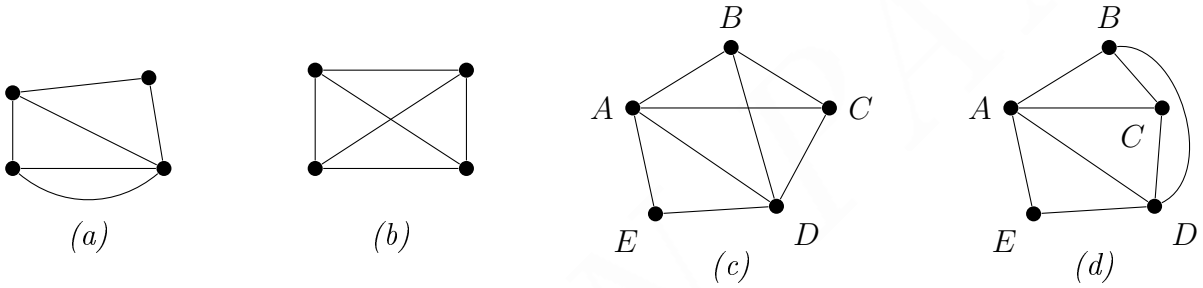
Planar & Dual Graphs

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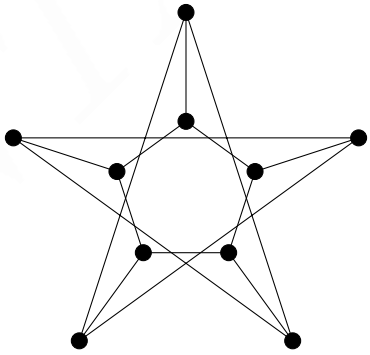
1 Planar Graph

Definition 1.1. *A graph is called plane graph if it can be drawn on a plane in such a way that any two of its edges either meet only at their end vertices or do not meet at all.*
A graph which is isomorphic to a plane graph is called planar graph.

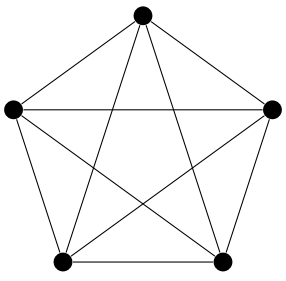
Example 1.2. *In the following graphs, graph (a) is a plane graph, graph (b) is isomorphic to the graph (a), so (b) is a planar graph. Graph (c) is also planar, as it is isomorphic to (d).*



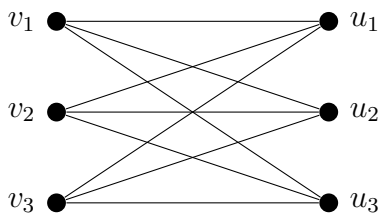
Definition 1.3. *A graph which is not planar, is called non-planar graph. The following graph can not be redraw without crossing over the edges. So this is a non-planar graph.*



Definition 1.4 (Kuratowski's First Graph)**.** *A complete graph with 5 vertices (K_5) is known as Kuratowski's first graph.*



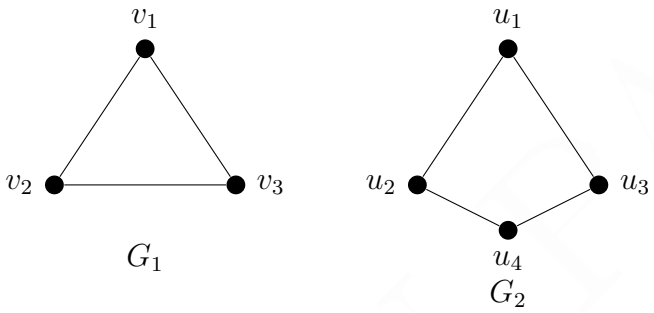
Definition 1.5 (Kuratowski's second graph)**.** *A regular connected graph with 6 vertices and 9 edges ($K_{3,3}$) is called Kuratowski's second graph.*



Definition 1.6 (Homeomorphism). *Two graphs G_1 and G_2 are said to be homeomorphic if one can be obtained from the other by subdividing edges (inserting vertices of degree 2 on edges) or by suppressing vertices of degree 2.*

In simple words, two graphs are homeomorphic if they have the same structure after adding or removing intermediate vertices on edges.

Example 1.7. *Consider the following two graphs:*



In the graph G_2, the edge v_2v_3 of G_1 is subdivided by inserting a new vertex u_4. If we remove the vertex u_4 and join u_2 directly to u_3, we obtain the graph G_1.

Hence, G_1 and G_2 are ***homeomorphic graphs****.*

Theorem 1.8 (Euler's Formula). *A connected planar graph G with n-vertices and e-number of edges determines $f = e - n + 2$ number of regions(faces).*

Theorem 1.9 (Generalized Euler's Formula). *A planar graph G with n-vertices and e-number of edges and k-number of connected components determines $f = e - n + k + 1$ number of regions(faces).*

Theorem 1.10. *Let G be a simple connected planar graph with n vertices, e edges and f regions (faces). Then*

(i) $e \geq \dfrac{3f}{2}$

(ii) $e \leq 3n - 6$

Proof. (i) If $n = 3$, then G may have 2 or 3 edges. If $e = 3$, then G has 2 faces, and if $e = 2$, then G has 1 face. Thus the inequality in (i) holds.

Now assume $n \geq 4$.

First consider the case when G is a tree. Then, $e = n - 1$, $\quad f = 1$.

Since a tree has no circuit and hence encloses no finite region, therefore, $e = n - 1 \geq 3$ and thus $e \geq \frac{3}{2} = \frac{3f}{2}$. Hence proof is done.

Now suppose G is not a tree. Then G contains at least one circuit, and every face is bounded by at least three edges because G has no loops or parallel edges.

Hence the number of boundary edges of each region is at least 3. Therefore

$$\text{(sum of boundary edges of all regions)} \geq 3f. \tag{1}$$

However, each edge is counted at most twice in this sum (once for each adjacent region). Hence

$$(\text{sum of boundary edges of all regions}) \leq 2e.$$

Combining with (1), we obtain $2e \geq 3f \quad \Rightarrow \quad e \geq \frac{3f}{2}$. Hence, proof is complete.

(ii) From Euler's formula for planar graphs, $f = e - n + 2$.

Substituting this into (i), we get $e \geq \frac{3(e-n+2)}{2}$, which implies, $e \leq 3n - 6$. □

Theorem 1.11. *Kuratowski's first graph K_5 is non-planar.*

Proof. Here, number of vertices(n)=5. As, sum of degrees of all vertices is twice the number of edges(e), so $4n = 2e$, which implies $e = 10$. But, $3n - 6 = 9$. This shows that, $e > 3n - 6$. SO the graph is non-planar, by above theorem. □

Example 1.12. *Show that, removing an edge from Kuratowski's first graph, makes the graph planar.*

Example 1.13. *Let G be a simple planar graph with fewer than 12 vertices. Prove that G has a vertex whose degree is ≤ 4.*

Solution. Let G have n vertices and e edges. Then, $n < 12$. Let $v_1, v_2, \cdots, v_n$ be the vertices of G.

If possible assume that, $\deg(v_i) \geq 5 \quad$ for all $i = 1, 2, \cdots, n$.

Then, $\sum_{i=1}^{n} \deg(v_i) \geq 5n$.

But we know that $\sum_{i=1}^{n} \deg(v_i) = 2e$.

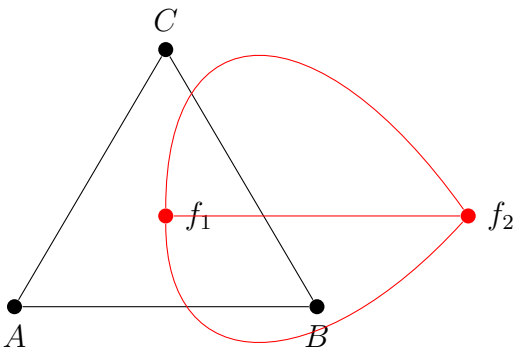
Hence, $2e \geq 5n$.

Now, for a simple planar graph, $e \leq 3n - 6$, implies $5n \leq 6n - 12$, implies $n \geq 12$.

This contradicts the assumption that $n < 12$. Therefore our assumption is false, and G must contain at least one vertex v such that $\deg(v) \leq 4$.

2 Dual Graph

Definition 2.1. *Let G be a planar graph. The* dual graph G^* *is obtained by placing one vertex in each face of G and joining two vertices by an edge whenever their corresponding faces in G share a common edge.*



In the above graph, black triangle is original planar graph G. There are two faces: inside and outside. Red vertices represent those faces. Three red edges corresponds to the three edges of the triangle. So, the dual graph G^* is the red graph (2-vertices and 3-parallel edges).